

Nokia Asha S40 series

Asha messenger phones are expected by late 2011/early 2012, featuring Nokia's improved browser for speedy browsing at low bandwidth

Nvidia Tegra 3 outed

Nvidia has launched its five-core Tegra 3 mobile chipset, featuring 5x the performance of Tegra 2

INTERVIEW

Behind the scenes, with a Ferrari electronics wizard

During the Formula 1 Indian Grand Prix race weekend, we caught up with Andrea Beneventi, Head of Electronics for Track and Test, who spoke in detail about the technological side of Formula 1. Here's what he had to say.

d Tell us more about what the Electronics for Track and Test department does in F1?

Andrea Beneventi: We essentially look after two things - the electronics development and technology scouting. Looking for new technologies that are applied to our system and what can be used in another way on the Formula 1 car. We're also developing components, and every year we send a number of components for different installations in the search of more accuracy, in search of lighter components. So, we buy these components as they are, and modify it for the Formula 1 environment. Everyone uses these components in a different environment. For an extreme example, we can put an accelerometer on the engine while someone else is using it on the chassis, and you can understand it's a completely different environment for vibrations and it means that I have to take a component used on another car and validate it for installation on our car. Obviously, if we find a new technology that could be applied to an F1 car or to an area of F1, we give it to the R&D team.

The other area is the application area, for all the departments where electronics is used inside Ferrari. In particular, there's the race, there's the test, there are test benches for engine, gearbox hydraulic and also whenever the knowledge of the electronics department has to be given inside Ferrari as well as at the track where we provide support to our client teams - Sauber and Toro Rosso - also.

We also manage the KERS at the track. We don't, however, make the project; there's a specific department that does that. The KERS is given to us and we manage it at the track, for tuning and logistics.

There's the development process we look at, and then we're asked to look for new technologies knowing the area where they have to be applied. We have 12 engineers and 25 people in total to manage things on the track.

d The boundaries and parameters in F1 have become tighter and more controlled. In that scenario, how do you differentiate and innovate to gain an advantage over rivals?

AB: Possibilities in Electronics, in particular, are being quite reduced in the area



Andrea Beneventi, Head of Electronics for Track and Test, Head of Support for Electronic Applications @Scuderia Ferrari

of development. Because from the starting of 2008, a single ECU (Engine Control Unit) and signal system has been introduced by the FIA. This means that the main ECU, the software that is running on the ECU that manages the performance of the car or engine or gearbox or feedback to the drivers is something that's fixed. The hypothesis of this chapter is that the ECU must be able to control the car you have and not force you to have particular mechanical solution for your needs. Providing that would be just the management of your mechanics and it's not that you can do whatever you want with the software. In this area, a lot of work we did before is fixed, but, we now have more time to concentrate on other areas - managing the performance of the car. Another constraint over the last few

years is the cost. In the last few years by agreement, we have what's called a "FOTA (Formula One Teams Association) budget", which is an agreement between the teams to have a maximum limit on the cost. An external company audits this. This has to be taken as a challenge because we cannot sacrifice performance. With a big cost reduction, we're talking about one-fourth of the budget we had six years ago, which is a big challenge. It's also a good challenge for us, technology wise. Sometimes, in order to reduce the cost, we have to jump to another technology. The reduction in budget isn't reducing the technology or the level of technology in the car. Compared to the car we had 10 years ago, the car now is technology-wise better.

d How does "telemetry" work? How is the data received from the car?

AB: There's a lot of data that's captured in the car, and downloaded from it. It's useful when we're changing the parameters. All the data that we have, that we receive from the car via the antennas on the car, are on the system provided by the FIA, so it's the same for all cars. The telemetry is not restricted. The data that is downloaded in real-time from the cars and the garage is stored on our server.

d How much data does the car usually send out? How is that data deciphered?

AB: We get about 2 GB to 3 GB of data from each car over a race weekend. That data is collected in different ways. All data from the track is synchronised with the server in Maranello (in Northern Italy) via the data connection with bandwidth of about 6 Mbps. The data is then organised in various forms. We send the data from the track to Maranello where it's collected and stored. The setup in Maranello is an extension of our on-track system. The need for this arises because of logistics. The systems that we carry with us are powerful and light, but limited. Over there, for example, we can use more powerful desktop PCs, whereas on the track, we have limited machines, usually laptops.